Richard Zhuang

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EDUCATION

University of California, Berkeley — Expected Graduation: Jun. 2025
Double Major in Applied Mathematics and Computer Science (Honor) — *GPA:* ***4.0/4.0***

Relevant Coursework: Machine Learning, Deep Learning, Foundations of Large Language Models (LLMs), Statistical Learning Theory, Probability Theory and Stochastic Processes, Techniques of Data Science, Computer Algorithms.

ACADEMIC PROJECTS

Learning Compact Representations of LLMs (In Proceedings of ICLR 2025) — Jan. 2024 - Nov. 2024
Advised by Prof. Jiantao Jiao

- Designed and implemented a **encoder-decoder based reconstruction system** to predict correctness of LLM responses on unseen queries and extract model embeddings.
- Conducted probing experiments to analyze information contained in the learned embeddings.

Training LLMs for Imperfect Information Games (In Proceedings of AAAI 2025) — Jan. 2024 - Sep. 2024
Advised by Prof. Gopala Anumanchipalli

- Established a **benchmarking dataset** with over 20K test and 560K training examples of game scenarios utilizing optimal decision labels verified by poker strategy solvers.
- Performed **supervised fine-tuning** on Llama-3-8B, achieving a 52% improvement in test set accuracy.

Multi-LLM-Agent Behavior Analysis (Published in ICML 2024) — Sep. 2023 - Feb. 2024
Advised by Prof. Dawn Song

- Conducted prompt engineering and performed simulations for the Prisoners Dilemma and Story Relay games between multiple LLM agents.
- Analyzed game results through data visualization to discern behavior disparities between multi-agent and single-agent systems.

EXPERIENCES

CMU Delphi Group — May. 2024 - Sep. 2024
Software Development Engineer Intern

- Contributed to real-time epidemiology forecasting tools by developing and alpha-testing R packages.
- Enhanced package documentations by establishing tutorial notebooks on time-series modeling using **ARIMA**, **ETS**, and **Fourier Decomposition methods**.

UCSF Mindscape Lab — Sep. 2023 - Sep. 2024
Machine Learning Researcher Intern

- Implemented a **Bayesian hierarchical generative model** of patient movement patterns to aid the development of a disease transmission simulation system.
- Adapted image reconstruction algorithms like **Masked Autoencoder** and **Vision Transformer** to recover omniscent transmission pattern from partial observations.

Sports Analytics Group at Berkeley — Aug. 2022 - May. 2024
Data Analyst

- Fall 2022: Developed a **decision tree regression model** to assess the fair market value of free agent players for the Los Angeles Sparks in WNBA.
- Fall 2023: Utilized **similarity scoring** and **clustering algorithms** to produce a scouting algorithm that identify cost-effective players for the Minnesota Lynx in WNBA.

SKILLS

- Languages: Python, Java, R, SQL, MATLAB, JavaScript
- Tools & Packages: PyTorch, HuggingFace, vLLM, LitGPT, Ray, NumPy, Pandas, Matplotlib, AWS